

UE Apprentissage SIA S9

Supervised Learning

Mathieu FAUVEL

CESBIO, Université de Toulouse, CNES/CNRS/INRAe/IRD/UPS, Toulouse, FRANCE

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Outline

Introduction

Introductory Example

- Linear model for regression

- Linear model for classification

Multi layer Perceptron

- Going deeper

- Needs some regularization

Convolutional Neural Networks

- Concepts

- How to improve the learning process

Recurrent Neural Networks

- When order matters

- Modern RNN

Attention Mechanisms

- Sequence of varying length

- Attention Score

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- Contact: mathieu.fauvel@inrae.fr
- CV:

2004-2007 Ph.D. degree in Signal and Image Processing from the INP, Grenoble & the University of Iceland

2007-2008 Assistant Professor Grenoble

2008-2010 Post-doc position at INRIA - MISTIS Team

2010-2011 Assistant Professor Toulouse

2011-2018 Associate Professor at DYNAFOR & INP, Toulouse

Since 2018 Research (CRCN) at CESBIO, INRAe

- Research interests are:

Machine learning for environmental/ecological monitoring

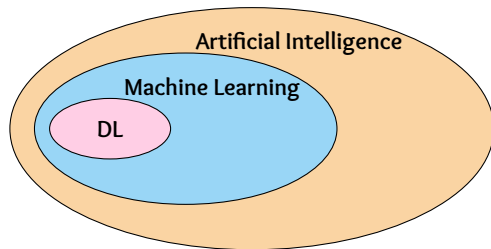
Course Objectives

- Learn modern supervised machine learning
- Understand how each step works
 - ★ Data préparation
 - ★ Model definition
 - ★ Optimization step
- Implement various Deep Learning models in PyTorch
- Application to Computer Vision, Remote Sensing, Time Series

What is Supervised (Machine) Learning ?

- **Artificial Intelligence**

Perform human tasks using computers and algorithms



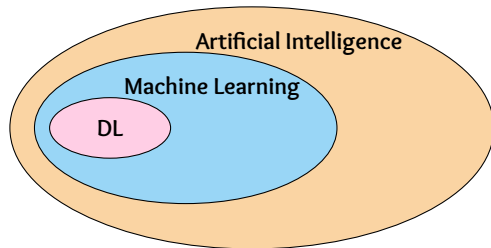
What is Supervised (Machine) Learning ?

- **Artificial Intelligence**

Perform human tasks using computers and algorithms

- Mitchell, *The discipline of machine learning:*

Machine Learning is defined as the capacity of a computer program to improve its performance measure with observations



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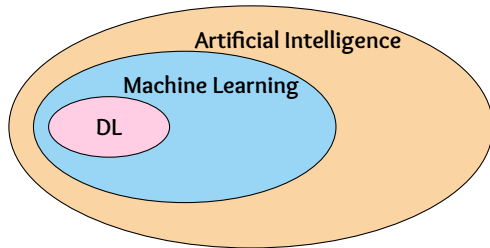
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- **Deep Learning**

- ★ Lot of data data
- ★ Complex model
- ★ High performance computing



Main Notations

Data

- Observed data $\mathbf{x} \in \mathbb{R}^d$, called *input variables* or *predictors*.
- Data to be predicted $y \in \mathbb{R}^p$, called *output variables* or *responses*.

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Learning Problem

- If y are quantitative data: *regression* and $y \in \mathbb{R}^p$.
- If y are categorical data: *classification* and $y \in \{C_1, \dots, C_C\}$ with C_i refers to the i^{th} category.

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Prediction function

$$\begin{aligned} f_{\theta} : \mathbb{R}^d &\rightarrow \mathbb{R}^p \\ \mathbf{x} &\mapsto y \end{aligned}$$

Online References

- Aston Zhang et al. *Dive into Deep Learning*. <https://D2L.ai>. Cambridge University Press, 2023
- Simon J.D. Prince. *Understanding Deep Learning*. MIT Press, 2023. URL: <https://udlbook.github.io/udlbook/>
- Sebastian Raschka, Yuxi (Hayden) Liu, and Vahid Mirjalili. *Machine Learning with PyTorch and Scikit-Learn*. Birmingham, UK: Packt Publishing, 2022. ISBN: 978-1801819312
- Kevin P. Murphy. *Probabilistic Machine Learning: An introduction*. MIT Press, 2022. URL: <http://probml.github.io/book1>
- Kevin P. Murphy. *Probabilistic Machine Learning: Advanced Topics*. MIT Press, 2023. URL: <http://probml.github.io/book2>

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Univariate linear model

- $f(x) = wx + b$ and $\theta = (w, b)$
- Loss function: $\ell(f(x_i), y_i) = (f(x_i) - y_i)^2$
- Objective function:

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n \ell_i$$

- Gradients update: $\theta^{t+1} = \theta^t - \eta \nabla_{\theta} \mathcal{L}$

$$\frac{\partial \mathcal{L}}{\partial w} = \frac{2}{n} \sum_{i=1}^n x_i (wx_i + b - y_i)$$

$$\frac{\partial \mathcal{L}}{\partial b} = \frac{2}{n} \sum_{i=1}^n (wx_i + b - y_i)$$

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Work

Implement the 1D regression in pytorch of the following function:

$$f(x) = 2x - 1$$

Do the notebook:

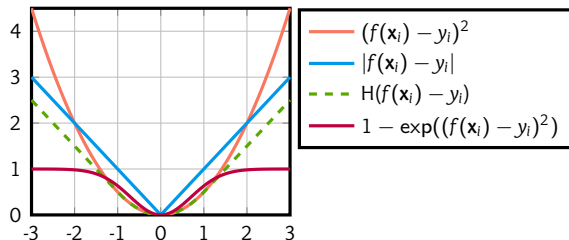
[regression_1d_toy.ipynb](#)

Multivariate linear model

- $f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$ and $\boldsymbol{\theta} = (\mathbf{w}, b)$
- Same loss and objective function and so same gradient updates ...

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- We can use other loss function

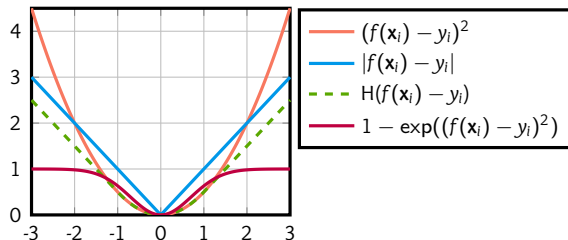


with H stands for *Hubert loss*:

$$H(f(\mathbf{x}_i) - y_i) = \begin{cases} \frac{1}{2} (f(\mathbf{x}_i) - y_i)^2 & \text{for } |f(\mathbf{x}_i) - y_i| \leq 1, \\ (|f(\mathbf{x}_i) - y_i| - 0.5), & \text{otherwise.} \end{cases}$$

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Work

On the **California housing** data set

- Prepare (train/validation) data
- Define a multi. linear model
- Optimize your hyperparameters (batch size, learning rate)
- Switch L2 to other loss function

Scaling feature for multivariate linear model with L2 loss function

- Suppose we have two features of different scale (e.g. because of different unit)

$$\mathbf{x}_1 \sim \mathcal{N}(0, 1) \text{ and } \mathbf{x}_2 \sim \mathcal{N}(10, 10)$$

- What is the impact on the gradient update ?

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- Noting $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n]^\top$ and $\mathbf{y} = [y_1, \dots, y_n]^\top$

$$\nabla_{\mathbf{w}} = -\mathbf{X}^\top \underbrace{(\mathbf{y} - \mathbf{X}\mathbf{w})}_{\mathbf{e}} \Rightarrow \nabla_{\mathbf{w}_p} = -\sum_{i=1}^n \mathbf{e}_i \mathbf{x}_{ip}$$

- Parameter update

$$\mathbf{w}_p^{(t+1)} = \mathbf{w}_p^{(t)} + \frac{\eta}{n} \sum_{i=1}^n \mathbf{e}_i^{(t)} \mathbf{x}_{ip}$$

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- Standardization of the input feature (and for regression the output too): [scaling_feature.ipynb](#)

$$\tilde{\mathbf{x}} = (\mathbf{x} - \boldsymbol{\mu}) \oslash \boldsymbol{\sigma}$$

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Classification as a regression problem

- Predict categorical variables ("Dogs", "Cats", "Monkey" ...)
- Estimation of the posterior probability

$$p(y_i = c | \mathbf{x}_i) \forall c \in \{1, \dots, C\} \text{ with } p(y_i = c | \mathbf{x}_i) \geq 0 \text{ and } \sum_{c=1}^C p(y_i = c | \mathbf{x}_i) = 1$$

- MAP rule: $\hat{c}_i = \arg \max_c p(y_i = c | \mathbf{x}_i)$

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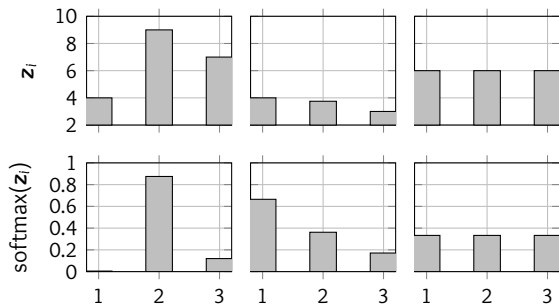
- MAP rule: $\hat{c}_i = \arg \max_c p(y_i = c | \mathbf{x}_i)$
- Softmax linear regression

$$1 = \sum_{c=1}^C p(y_i = c | \mathbf{x}_i) = \sum_{c=1}^C \exp \left\{ \mathbf{w}_c^\top \mathbf{x}_i + b_c + K_i \right\} = \exp \{ K_i \} \sum_{c=1}^C \exp \left\{ \mathbf{w}_c^\top \mathbf{x}_i + b_c \right\}$$

thus

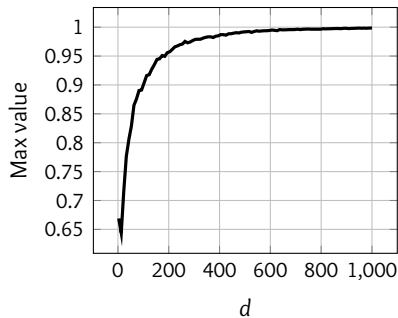
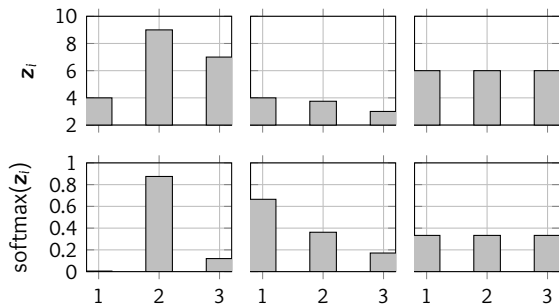
$$p(y_i = c | \mathbf{x}_i) = \frac{\exp \left\{ \mathbf{w}_c^\top \mathbf{x}_i + b_c \right\}}{\sum_{c'=1}^C \exp \left\{ \mathbf{w}_{c'}^\top \mathbf{x}_i + b_{c'} \right\}} = \frac{\exp(\mathbf{z}_{ic})}{\sum_{c'=1}^C \exp(\mathbf{z}_{ic'})} = \text{softmax}(\mathbf{z}_i)_c$$

Softmax



- Translation invariant: $\text{softmax}(\mathbf{z} + K) = \text{softmax}(\mathbf{z})$
- Preserve ordering relation: $\arg \max_c(\mathbf{p}_i) = \arg \max_c(\mathbf{z}_i)$
- If $\mathbf{z} \in \mathbb{R}^d$ with $d \rightarrow \infty$ then $\text{softmax}(\mathbf{z}) \rightarrow \text{"One-hot-vector"}$

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- If $\mathbf{z} \in \mathbb{R}^d$ with $d \rightarrow \infty$ then $\text{softmax}(\mathbf{z}) \rightarrow \text{"One-hot-vector"}$

Cross entropy loss function

- One hot encoding: $y_i = c \rightarrow \mathbf{y}_i \in \mathbb{R}^C$, $y_{ic} = 1$ if $i = c$ else 0
- Likelihood:

$$l(\mathbf{y}_i, \hat{\mathbf{y}}_i) = \prod_{c'=1}^C p(y_i = c' | \mathbf{x}_i)^{y_{ic'}} = \frac{\exp(\mathbf{z}_{ic})}{\sum_{c'=1}^C \exp(\mathbf{z}_{ic'})}$$

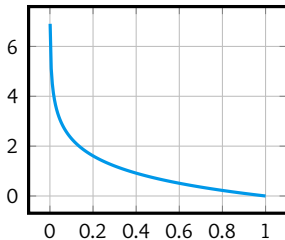
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- Negative log likelihood - **Cross entropy**:

$$\begin{aligned}\ell(\mathbf{y}_i, \hat{\mathbf{y}}_i) &= - \sum_{c'=1}^C y_{ic'} \log \{p(y_i = c' | \mathbf{x}_i)\} \\ &= - \log \{p(y_i = c | \mathbf{x}_i)\} \\ &= - (\mathbf{w}_c^\top \mathbf{x}_i + b_c) + \underbrace{\log \left\{ \sum_{c'=1}^C \exp(\mathbf{w}_{c'}^\top \mathbf{x}_i + b_{c'}) \right\}}_{\text{logsumexp}(\mathbf{z}_i)}\end{aligned}$$



Discussion

What happens if:

- All $\mathbf{z}_{ic}$ are small ?
- One $\mathbf{z}_{ic}$ is very large ?
- All $\mathbf{z}_{ic}$ are very large ?

Multivariate linear classifier

- The model

$$\mathbf{p}_i = \text{softmax}(\mathbf{W}\mathbf{x}_i + \mathbf{b})$$

- Cross entropy loss function
- Same comments than for regression
 - ★ Split train, val and test
 - ★ Standardization
 - ★ Batch training
 - ★ Tune learning rate

Work

Implement a linear classifier in pytorch for the classification of **Fashion MNIST** data set.

- Monitor

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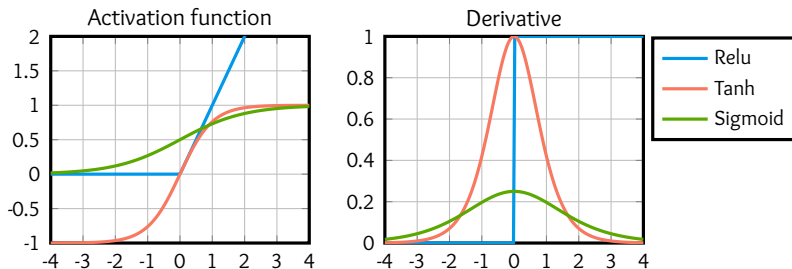
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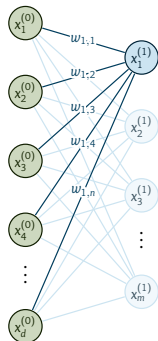
Activation function

- As discussed earlier we need non-linearity between two layers
- Done with *activation function* between two layers
 - ★ Rectified Linear (ReLU): $\max(x, 0)$
 - ★ Sigmoid: $\frac{1}{1 + \exp(-x)}$
 - ★ Tanh: $\tanh(x)$



Multi Layer Perceptron (MLP)

- $\mathbf{x}^{(1)} = \sigma(\mathbf{W}^{(0)}\mathbf{x}^{(0)} + \mathbf{b}^{(0)})$
- $\mathbf{x}^{(2)} = \sigma(\mathbf{W}^{(1)}\mathbf{x}^{(1)} + \mathbf{b}^{(1)})$
- ...
- $\mathbf{y} = \mathbf{W}^H\mathbf{x}^H + \mathbf{b}^H$



$$= \sigma(w_{1,0}x_0^{(0)} + w_{1,1}x_1^{(0)} + \dots + w_{1,d}x_d^{(0)} + b_1^{(0)})$$

$$= \sigma\left(\sum_{i=1}^d w_{1,i}x_i^{(0)} + b_1^{(0)}\right)$$

$$\begin{pmatrix} x_1^{(1)} \\ x_2^{(1)} \\ \vdots \\ x_m^{(1)} \end{pmatrix} = \sigma \left[\begin{pmatrix} w_{1,0} & w_{1,1} & \dots & w_{1,d} \\ w_{2,0} & w_{2,1} & \dots & w_{2,d} \\ \vdots & \vdots & \ddots & \vdots \\ w_{m,0} & w_{m,1} & \dots & w_{m,d} \end{pmatrix} \begin{pmatrix} x_1^{(0)} \\ x_2^{(0)} \\ \vdots \\ x_d^{(0)} \end{pmatrix} + \begin{pmatrix} b_1^{(0)} \\ b_2^{(0)} \\ \vdots \\ b_m^{(0)} \end{pmatrix} \right]$$

$$\mathbf{x}^{(1)} = \sigma(\mathbf{W}^{(0)}\mathbf{x}^{(0)} + \mathbf{b}^{(0)})$$

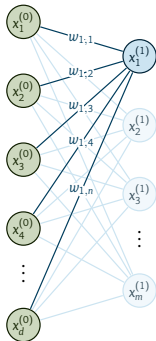
From https://tikz.net/neural_networks/

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- ...
- $\mathbf{y} = \mathbf{W}^H\mathbf{x}^H + \mathbf{b}^H$

Work

- Why the last layer does not have non-linearity?
- For an univariate regression problem, shows that a 2-layers MLP is a piece-wise linear function.
- Implement an MLP classifier for Fashion MNIST data set



$$\begin{aligned} &= \sigma(w_{1,0}x_0^{(0)} + w_{1,1}x_1^{(0)} + \dots + w_{1,d}x_d^{(0)} + b_1^{(0)}) \\ &= \sigma\left(\sum_{l=1}^d w_{1,l}x_l^{(0)} + b_1^{(0)}\right) \end{aligned}$$

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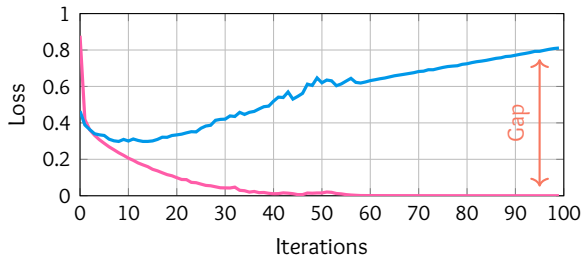
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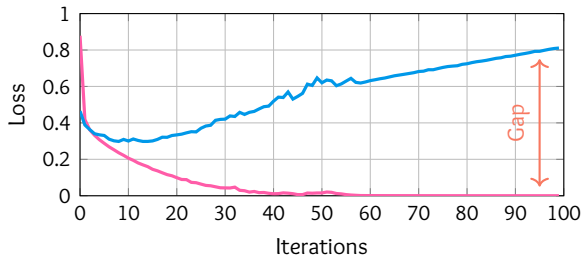
Generalization Error

- *Generalization error*: Difference between the train accuracy and val/test accuracy
- *Overfitting*: High generalization error



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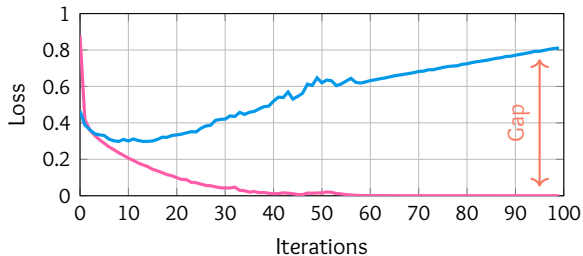
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Theory: favor simpler model and/or smooth model

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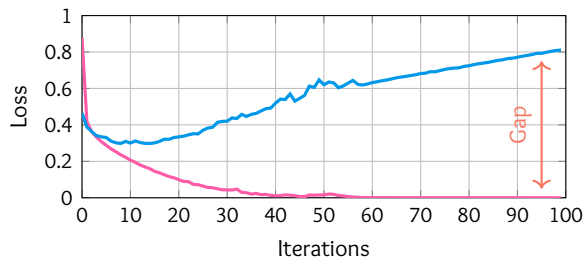


Theory: favor simpler model and/or smooth model

- Early stopping: monitor the validation loss

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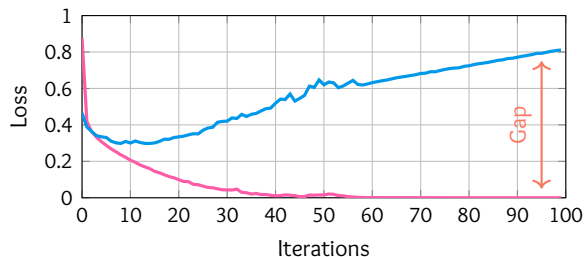
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Theory: favor simpler model and/or smooth model

- Early stopping: monitor the validation loss
- Weight decay: Tikhonov regularization
- Noise injection: Dropout

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n \ell_i + \frac{\lambda}{H} \sum_{h=1}^H \|\mathbf{w}_h\|_2^2$$

Dropout

- Smoothness: $f(\mathbf{x} + \epsilon) \approx f(\mathbf{x})$
- Idea: Noise injection during the learning process such as $\mathbb{E}[\tilde{\mathbf{x}}] = \mathbf{x}$
- With MLP:

$$f(\mathbf{x}) = f_H \circ f_{H-1} \circ \dots \circ f_0(\mathbf{x})$$

$$\text{and } \mathbf{x}^{(h+1)} = f_h(\mathbf{x}^{(h)}) = \sigma(\mathbf{W}^{(h)}\mathbf{x}^{(h)} + \mathbf{b}^{(h)})$$

- Dropout: remove (set to zero) some internal features with probability p

$$\mathbf{x}^{(h+1)} = f_h(\tilde{\mathbf{x}}^{(h)}) = \sigma(\mathbf{W}^{(h)}(\mathbf{m}^{(h)} \odot \mathbf{x}^{(h)}) + \mathbf{b}^{(h)})$$

with

$$\mathbf{m}_i^{(h)} = \begin{cases} 0 & \text{with probability } p \\ 1/(1-p) & \text{with probability } 1-p \end{cases}$$

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- Idea: Noise injection during the learning process such as $\mathbb{E}[\tilde{\mathbf{x}}] = \mathbf{x}$
- With MLP:

$$f(\mathbf{x}) = f_H \circ f_{H-1} \circ \dots \circ f_0(\mathbf{x})$$

$$\text{and } \mathbf{x}^{(h+1)} = f_h(\mathbf{x}^{(h)}) = \sigma(\mathbf{W}^{(h)}\mathbf{x}^{(h)} + \mathbf{b}^{(h)})$$

- Dropout: remove (set to zero) some internal features with probability p

$$\mathbf{x}^{(h+1)} = f_h(\tilde{\mathbf{x}}^{(h)}) = \sigma(\mathbf{W}^{(h)}(\mathbf{m}^{(h)} \odot \mathbf{x}^{(h)}) + \mathbf{b}^{(h)})$$

with

$$\mathbf{m}_i^{(h)} = \begin{cases} 0 & \text{with probability } p \\ 1/(1-p) & \text{with probability } 1-p \end{cases}$$

- Enable during training and disable during inference (can be used to estimate posterior distribution)

$$f(\tilde{\mathbf{x}}) = f_H \circ D_H \circ f_{H-1} \circ \dots \circ D_1 \circ f_0(\mathbf{x})$$

Work

- Implement early stopping
- Implement dropout
- (Optional) Implement Tikhonov regularization (weight decay in pytorch)

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- How to improve the learning process

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- Modern RNN

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- Sequence of varying length

- Attention Score

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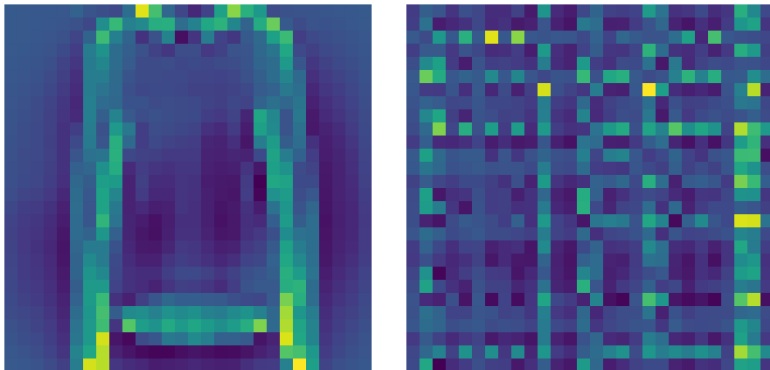
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An image is not a vector of pixels



2D Convolution

- Fully connected layer:

$$x_{pq}^{(1)} = \sigma \left(\sum_{i,j=1}^{H,W} \mathbf{w}_{pqij}^{(0)} x_{ij}^{(0)} + b_{pq}^{(0)} \right)$$

- Act locally: restrict to neighborhood of pixel p, q

$$x_{pq}^{(1)} = \sigma \left(\sum_{i,j=-\Delta}^{i,j=+\Delta} \mathbf{w}_{ij}^{(0)} x_{(p+i)(q+j)}^{(0)} + b^{(0)} \right)$$

$$\begin{pmatrix} w_{pq11} & w_{pq12} & \dots & w_{pq1j} & \dots & w_{pq1W} \\ w_{pq21} & w_{pq22} & \dots & w_{pq2j} & \dots & w_{pq2W} \\ \vdots & \vdots & & \vdots & & \vdots \\ w_{pqi1} & w_{pqi2} & \dots & w_{pqij} & \dots & w_{pqiW} \\ \vdots & \vdots & & \vdots & & \vdots \\ w_{pqH1} & w_{pqH2} & \dots & w_{pqHj} & \dots & w_{pqHW} \end{pmatrix}$$

2D Convolution

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$$\begin{pmatrix} 0 & \dots & 0 & \dots & 0 \\ 0 & \dots & 0 & \dots & 0 \\ \vdots & w_{(i)(j-1)} & w_{(i-1)(j)} & w_{(i-1)(j+1)} & \vdots \\ 0 & w_{(i)(j-1)} & w_{(i)(j)} & w_{(i)(j+1)} & 0 \\ \vdots & w_{(i+1)(j-1)} & w_{(i+1)(j)} & w_{(i+1)(j+1)} & \vdots \\ 0 & \dots & 0 & \dots & 0 \end{pmatrix}$$

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Work

Show that 2D convolution can be expressed as a linear layer.

Basics on 2D Convolution 1/2

Convolution

0	1	2	3	4
5	6	7	8	9
10	11	12	13	14
15	16	17	18	19

 $\chi^{(h)}$

1	1	1
1	1	1
1	1	1

 K
$$\begin{matrix} & * & \\ & & = \end{matrix}$$

54	63	72
99	108	117

 $\chi^{(h+1)}$

(More https://github.com/vdumoulin/conv_arithmetic/blob/master/README.md)

Basics on 2D Convolution 1/2

Convolution + Padding (1,1)

0	0	0	0	0	0	0
0	0	1	2	3	4	0
0	5	6	7	8	9	0
0	10	11	12	13	14	0
0	15	16	17	18	19	0
0	0	0	0	0	0	0

*

1	1	1
1	1	1
1	1	1

=

12	21	27	33	24
33	54	63	72	51
63	99	108	117	81
52	81	87	93	64

Basics on 2D Convolution 1/2

Convolution + Padding (1,1) + Stride (3, 2)

0	0	0	0	0	0	0
0	0	1	2	3	4	0
0	5	6	7	8	9	0
0	10	11	12	13	14	0
0	15	16	17	18	19	0
0	0	0	0	0	0	0

*

1	1	1
1	1	1
1	1	1

=

12	27	24
52	87	64

Basics on 2D Convolution 1/2

Convolution for multivalued images

0	1	2	3	4
5	6	7	8	9
10	11	12	13	14
15	16	17	18	19

*

1	1	1
1	1	1
1	1	1

=

54	63	72
99	108	117

+

=

288	279	270
243	234	225

19	18	17	16	15
14	13	12	11	10
9	8	7	6	5
4	3	2	1	0

*

2	2	2
2	2	2
2	2	2

=

234	216	198
144	126	108

Basics of 2D Convolution 2/2

- Multivariate input - Multivariate output

$$\mathbf{X}^{(h)} \in \mathbb{R}^{C_{(h)} \times H^{(h)} \times W^{(h)}}$$

$$\mathbf{K}^{(h)} \in \mathbb{R}^{C_{(h+1)} \times C_{(h)} \times k_H^h \times k_W^h}$$

$$\mathbf{X}^{(h+1)} \in \mathbb{R}^{C_{(h+1)} \times H^{(h+1)} \times W^{(h+1)}}$$

- Special case of 1×1 convolution: $(C_{(h+1)}, C_{(h)}, k_H, k_W) = (C_o, C_i, 1, 1)$
 - ★ Combine pixel across channels - no spatial operation
 - ★ Matrix product / linear layer in the channel dimension

$$\mathbf{X}^{(h+1)} = \text{reshape} \left\{ \sigma \left(\mathbf{K}^{(h)} \text{reshape} \{ \mathbf{X}^{(h)}, C_{(h)}, H^{(h)} W^{(h)} \} + \mathbf{b}^{(h)} \right), C_{(h+1)}, H^{(h)}, W^{(h)} \right\}$$

$$\text{with } \text{reshape} \{ \mathbf{X}^{(h)}, C_{(h)}, H^{(h)} W^{(h)} \} \in \mathbb{R}^{C_{(h)} \times H^{(h)} W^{(h)}}$$

Pooling

- Reduce the resolution of the image $\equiv$ Use larger convolution kernel
- Downsampling with simple operator on the neighborhood:
 - ★ Average value
 - ★ Max value (preferred)
- After the non-linearity

0	1	2	3	4
5	6	7	8	9
10	11	12	13	14
15	16	17	18	19

2×2 MaxPooling
Stride=2

6	8
16	18

Discussion

- Show that the successive convolution with two kernels is equivalent to one convolution.
- What stride size would you use for a max pooling with a kernel of size 2×2 ?

Work

Implement a CNN for the classification of *Fashion MNIST* data set: use one or two convolutional layers for the MLP.

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Batch Normalization

- Remember scaling problem in slide 10
- Similar problem may happen to each layer **but** we cannot use the same trick: the sample mean/variance change after each batch update !

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- Apply normalization per batch: *Batch Normalization*

$$\tilde{\mathbf{x}}_B = \gamma \odot \left[(\mathbf{x}_B - \mu_B) \oslash \sigma_B \right] + \beta$$

with μ_B and σ_B computed on batch B , and γ & β are learnable parameters

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Discussion

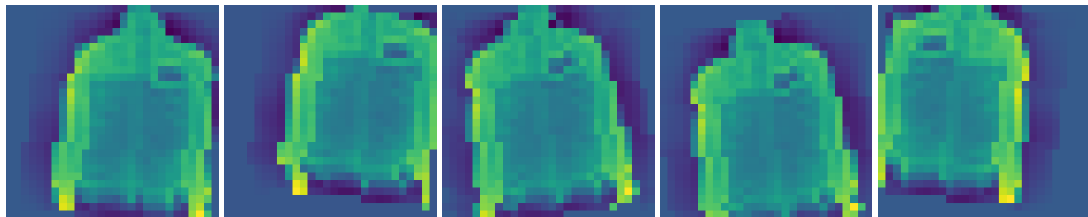
- What happen if the batch size is one ?
- What happen if the batch size is small ?
- What should we do in *prediction* mode ?

Work

Add normalization layer (2D) to your model.

Data Augmentation

- Noise injection to generate / augment data set: $(\mathbf{x}, y) \rightarrow (\tilde{\mathbf{x}}, y)$
- For image classification it is *relatively* easy:
 - ★ Geometric transformation: Rotation, translation, symmetry, radiometry ...
 - ★ Radiometric transformation: Invert, jitter ...
 - ★ *Any transformation that can model the variability of the acquisition process*



Work

Implement data augmentation with Pytorch for the training data only.

- Deep Neural Network: $f_h(\mathbf{x}^{(h)}) := f_h(\mathbf{x}^{(h)}, \theta_h)$

$$f(\mathbf{x}) = f_H \circ f_{H-1} \circ \dots \circ f_0(\mathbf{x})$$

Residual Network

- Deep Neural Network: $f_h(\mathbf{x}^{(h)}) := f_h(\mathbf{x}^{(h)}, \theta_h)$

$$f(\mathbf{x}) = f_H \circ f_{H-1} \circ \dots \circ f_0(\mathbf{x})$$

- Let $f_h = g_2 \circ g_1 \circ g_0$

$$\mathbf{x} \xrightarrow{g_0} \mathbf{x}^{(1)} \xrightarrow{g_1} \mathbf{x}^{(2)} \xrightarrow{g_2} f_h(\mathbf{x})$$

$$\frac{\partial f_h(\mathbf{x})}{\partial \theta_0} = \underbrace{\frac{g_2}{g_1} \circ \frac{g_1}{g_0}}_{\rightarrow} \frac{g_0}{\theta_0}(\mathbf{x}) \qquad \frac{\partial g(\mathbf{x})}{\partial \theta_1} = \frac{g_2}{g_1} \circ \frac{g_1}{\theta_1}(\mathbf{x}^{(1)}) \qquad \frac{\partial g(\mathbf{x})}{\partial \theta_2} = \frac{g_2}{\alpha_2}(\mathbf{x}^{(2)})$$

Residual Network

- Deep Neural Network: $f_h(\mathbf{x}^{(h)}) := f_h(\mathbf{x}^{(h)}, \theta_h)$

$$f(\mathbf{x}) = f_H \circ f_{H-1} \circ \dots \circ f_0(\mathbf{x})$$

- Let $f_h = g_2 \circ g_1 \circ g_0$ with *residual connection*

$$\mathbf{x} \xrightarrow{g_0} \oplus \rightarrow \mathbf{x}^{(1)} \xrightarrow{g_1} \oplus \rightarrow \mathbf{x}^{(2)} \xrightarrow{g_2} \oplus \rightarrow f_h(\mathbf{x})$$

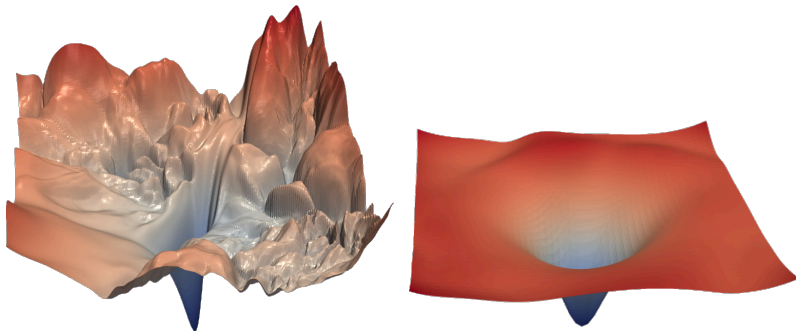
$$\frac{\partial f_h(\mathbf{x})}{\partial \alpha_0} = \frac{g_2}{g_1} \circ \frac{g_1}{g_0} \circ \frac{g_0}{\alpha_0}(\mathbf{x}) + \frac{g_1}{g_0} \circ \frac{g_0}{\alpha_0}(\mathbf{x}) + \frac{g_0}{\alpha_0}(\mathbf{x})$$

- ★ Improve gradient propagation
- ★ Prevent vanishing/shattered gradient

Residual Network

- Deep Neural Network: $f_h(\mathbf{x}^{(h)}) := f_h(\mathbf{x}^{(h)}, \theta_h)$

$$f(\mathbf{x}) = f_H \circ f_{H-1} \circ \dots \circ f_0(\mathbf{x})$$



From Hao Li et al. “Visualizing the Loss Landscape of Neural Nets”. In: *Proceedings of the 32nd International Conference on Neural Information Processing Systems*. NIPS’18. Montréal, Canada: Curran Associates Inc., 2018, pp. 6391–6401

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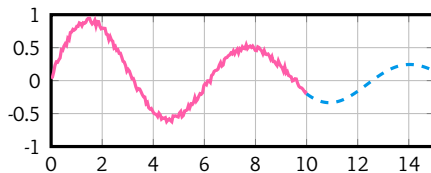
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Sequential data

- Time series



- Text data

This course is really cool

Auto-regressive model

- Autogressive models: $p(\mathbf{x}_t | \mathbf{x}_{t-1}, \dots, \mathbf{x}_1)$
- Sequence model:

$$p(\mathbf{x}_T, \mathbf{x}_{T-1}, \dots, \mathbf{x}_1) = \prod_{t=2}^T p(\mathbf{x}_t | \mathbf{x}_{t-1}, \dots, \mathbf{x}_1) p(\mathbf{x}_1)$$

- Markov model:

$$p(\mathbf{x}_T | \mathbf{x}_{T-1}, \dots, \mathbf{x}_1) = \prod_{t=2}^T p(\mathbf{x}_t | \mathbf{x}_{t-1}) p(\mathbf{x}_1)$$

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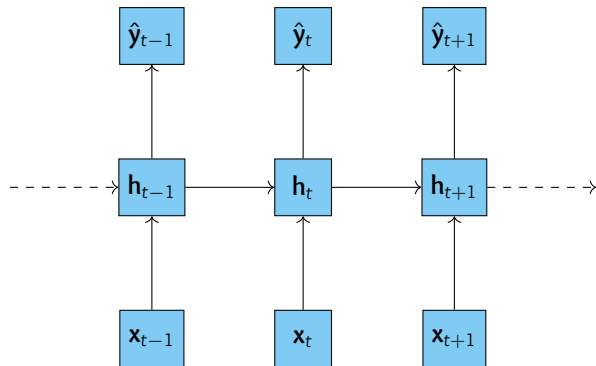
$$p(\mathbf{x}_T, \mathbf{x}_{T-1}, \dots, \mathbf{x}_1) = \prod_{t=2}^T p(\mathbf{x}_t | \mathbf{x}_{t-1}, \dots, \mathbf{x}_1) p(\mathbf{x}_1)$$

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$$p(\mathbf{x}_T | \mathbf{x}_{T-1}, \dots, \mathbf{x}_1) = \prod_{t=2}^T p(\mathbf{x}_t | \mathbf{x}_{t-1}) p(\mathbf{x}_1)$$

- Example: $p(\mathbf{x}_t | \mathbf{x}_{t-1})$ could model
 - ★ The probability to get letter after a observing a specific letter (or a set of)
 - ★ The expected value of a signal given past values of the same signal

Latent autoregressive model



- $\mathbf{h}_t = f(\mathbf{x}_t, \mathbf{h}_{t-1}) = \sigma_h(\mathbf{W}_{hx}\mathbf{x}_t + \mathbf{W}_{hh}\mathbf{h}_{t-1} + \mathbf{b}_h)$
- $\hat{\mathbf{y}}_t = g(\mathbf{h}_t) = \sigma_o(\mathbf{W}_{oh}\mathbf{h}_t + \mathbf{b}_o)$
- Backpropagation through time - BPTT

$$\ell = \frac{1}{T} \sum_{t=1}^T \ell(\mathbf{y}_t, \hat{\mathbf{y}}_t)$$

with $\mathbf{h}_0 = \mathbf{0}$

Issue with BPTT

$$\begin{aligned}\frac{\partial \ell}{\partial \mathbf{W}_{hh}} &= \frac{1}{T} \sum_{t=1}^T \frac{\partial \ell(\mathbf{y}_t, \hat{\mathbf{y}}_t)}{\partial \mathbf{W}_{hh}} \\&= \frac{1}{T} \sum_{t=1}^T \frac{\partial \ell(\mathbf{y}_t, \hat{\mathbf{y}}_t)}{\partial \hat{\mathbf{y}}_t} \frac{\partial \hat{\mathbf{y}}_t}{\partial \mathbf{W}_{hh}} \\&= \frac{1}{T} \sum_{t=1}^T \frac{\partial \ell(\mathbf{y}_t, \hat{\mathbf{y}}_t)}{\partial \hat{\mathbf{y}}_t} \frac{\partial g(\mathbf{h}_t)}{\partial \mathbf{h}_t} \left[\frac{\partial f(\mathbf{x}_t, \mathbf{h}_{t-1})}{\partial \mathbf{W}_{hh}} + \frac{f(\mathbf{x}_t, \mathbf{h}_{t-1})}{\partial \mathbf{h}_{t-1}} \frac{\partial \mathbf{h}_{t-1}}{\partial \mathbf{W}_{hh}} \right] \\&= \frac{1}{T} \sum_{t=1}^T \frac{\partial \ell(\mathbf{y}_t, \hat{\mathbf{y}}_t)}{\partial \hat{\mathbf{y}}_t} \frac{\partial g(\mathbf{h}_t)}{\partial \mathbf{h}_t} \left[\frac{\partial f(\mathbf{x}_t, \mathbf{h}_{t-1})}{\partial \mathbf{W}_{hh}} + \sum_{p=1}^{t-1} \left(\prod_{q=p+1}^t \frac{\partial f(\mathbf{x}_q, \mathbf{h}_{q-1})}{\partial \mathbf{h}_{q-1}} \right) \frac{\partial f(\mathbf{x}_p, \mathbf{h}_{p-1})}{\partial \mathbf{W}_{hh}} \right]\end{aligned}$$

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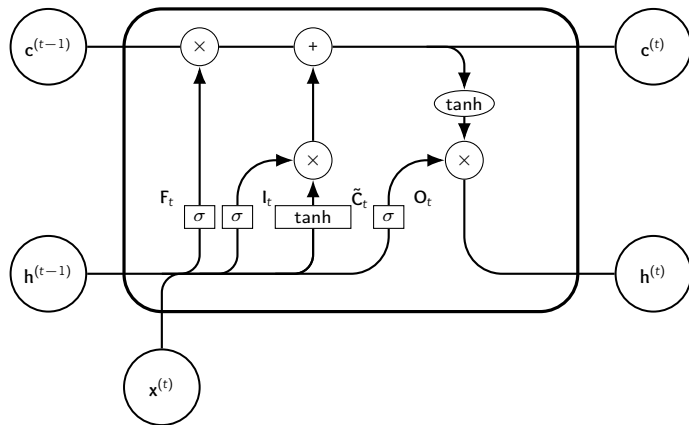
Modern RNN

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Long Short Term Memory (LSTM)



- Forget gate

$$F_t = \sigma(\mathbf{w}_{fh}h_{(t-1)} + \mathbf{w}_{fx}\mathbf{x}_t + \mathbf{b}_f)$$

- Input gate

$$I_t = \sigma(\mathbf{w}_{ih}h_{(t-1)} + \mathbf{w}_{ix}\mathbf{x}_t + \mathbf{b}_i)$$

- Input node

$$\tilde{C}_t = \tanh(\mathbf{w}_{\tilde{c}h}h_{(t-1)} + \mathbf{w}_{\tilde{c}x}\mathbf{x}_t + \mathbf{b}_{\tilde{c}})$$

- Output gate

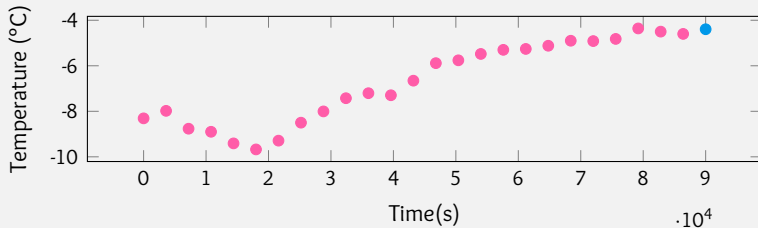
$$O_t = \sigma(\mathbf{w}_{oh}h_{(t-1)} + \mathbf{w}_{ox}\mathbf{x}_t + \mathbf{b}_o)$$

Good overview of LSTM: <https://colah.github.io/posts/2015-08-Understanding-LSTMs/>

Sequence prediction with RNN 1/2

Learning problem

Learn a function f that, given a fix sized sequence of measurements, predicts the most probable next one



Sequence prediction with RNN 2/2

- Regression problem on sequence,

$$\mathbf{x}_{T+1} = f(\mathbf{x}_T, \dots, \mathbf{x}_0)$$

- Then for longer predictions:

$$\mathbf{x}_{T+2} = f(\hat{\mathbf{x}}_{T+1}, \dots, \mathbf{x}_0)$$

Work

- Implement a RNN that learn to predict the next value of a sequence
- Once trained, generate one day of prediction (24 hours)

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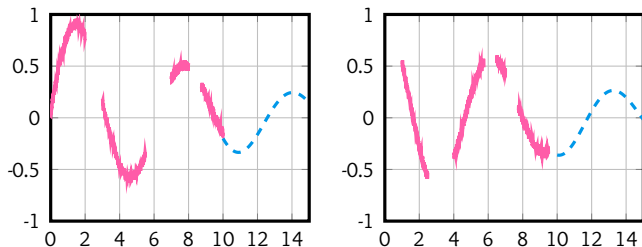
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Sequence are not always of same lenght

- Time series



- Text data

This course is really cool
Mathieu Fauvel course is really cool
Supervised Learning unit is so cool

Idea from database query

- Suppose a given data base made of m *(key, value)*¹:

$$\mathcal{DB} \stackrel{\text{def}}{=} \{(\mathbf{k}_1, \mathbf{v}_1), \dots, (\mathbf{k}_m, \mathbf{v}_m)\}$$

- Exact match for query $\mathbf{q} = \mathbf{k}_i$ return value $\mathbf{v}_i$, i.e.,

$$f(\mathbf{q}) = \mathbf{v}_i$$

¹Ideas gathered from <https://d2l.ai/>

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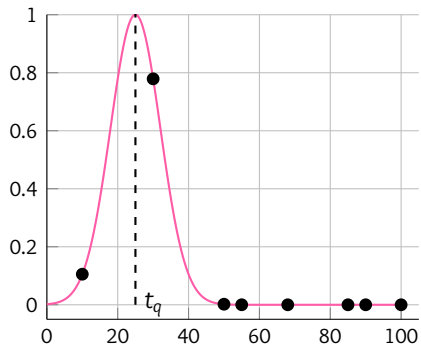
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- Approximate match for $\mathbf{q}$: Linear combination of values for *close* keys
How to compute/define *close* between $\mathbf{q}$ and $\mathbf{k}_i$?

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Example with time



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Dot product attention

- Scaled dot product

$$\alpha(\mathbf{q}, \mathbf{k}_i) = \frac{\exp(\mathbf{q}^\top \mathbf{k}_i / \sqrt{d})}{\sum_{j=1}^m \exp(\mathbf{q}^\top \mathbf{k}_j / \sqrt{d})}$$

$$f(\mathbf{q}) = \sum_{i=1}^m \alpha(\mathbf{q}, \mathbf{k}_i) \mathbf{v}_i$$

- If missing values: sum over $l < m$!
- Masked softmax:

$$\alpha = \text{softmax}(\mathbf{m} \otimes \mathbf{Kq})$$

with $\mathbf{m}_j = -\infty$ if key j is missing and 0 otherwise.

- Time Series:

- ★ $t_i \rightarrow \mathbf{k}_i$
- ★ $\mathbf{x}_i \rightarrow \mathbf{v}_i$
- ★ $t \rightarrow \mathbf{q}$

- In practice

- ★ Use embedding for t

$$t \mapsto \mathbf{e} = [\cos(2\pi\omega_1 t), \dots, \cos(2\pi\omega_E t), \\ \sin(2\pi\omega_1 t), \dots, \sin(2\pi\omega_E t)]$$

- ★ Then go through linear layer

$$\mathbf{k}_i = \mathbf{W}_k \mathbf{e}_i$$

$$\mathbf{v}_i = \mathbf{W}_v \mathbf{x}_i$$

MultiHeadAttention

- Use several attention heads

$$f_h(\mathbf{q}) = \sum_{i=1}^m \alpha(\mathbf{q}_h, \mathbf{k}_{hi}) \mathbf{v}_{hi}$$

and use an additional linear layer

$$f = \mathbf{W} \begin{bmatrix} f_1 \\ \vdots \\ f_H \end{bmatrix}$$

- Efficient implementation:

$$\mathbf{K} = [\mathbf{E}_1 \mid \dots \mid \mathbf{E}_H] [\mathbf{W}_{k1} \mid \dots \mid \mathbf{W}_{kH}] \in \mathbb{R}^{B \times (EH)}$$

Work

Implement time series interpolation using *Attention Mechanisms*.

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